Year 12 Maths - Quadratics

Similar questions - Completing the square

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The cross-section of an arch is modelled by the curve $y = -x^2 + 6x + 2$, where x and y are in metres and $y \geq 0$ is the height above the ground.

(a) Express $-x^2 + 6x + 2$ in the form $p - (x - q)^2$. **[2]**

(b) State the maximum height of the arch and the value of x at which it occurs. **[1]**

(c) Find the width of the arch at ground level, giving your answer as an exact surd. **[3]**

2 The function g is defined by $\mathrm{g}(x) = 3x^2 - 12x + 14$.

(a) Write $\mathrm{g}(x)$ in the form $a(x-b)^2 + c$. **[3]**

(b) Hence show that $\mathrm{g}(x) > 0$ for all values of x. **[1]**

(c) Describe a sequence of two transformations that maps the graph of $y = x^2$ onto the graph of $y = \mathrm{g}(x)$. **[3]**

Question 1

(a) $-x^2+6x+2 = -\left[(x-3)^2-9\right]+2 = 11-(x-3)^2$

(b) Maximum height 11 m, at $x = 3$

(c) $y = 0$: $(x-3)^2 = 11$, so $x = 3 \pm \sqrt{11}$. Width $= (3+\sqrt{11}) - (3-\sqrt{11}) = 2\sqrt{11}$ m

Question 2

(a) $3(x^2-4x)+14 = 3\left[(x-2)^2-4\right]+14 = 3(x-2)^2+2$

(b) $(x-2)^2 \geq 0$, so $g(x) \geq 2 > 0$

(c) A stretch parallel to the y-axis with scale factor 3 (giving $y = 3x^2$), followed by a translation of $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$